MANAV ARORA

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SUMMARY

Data scientist with 2+ years of experience shipping production machine learning in regulated financial environments at PwC, IEEE-published applied AI research at Envision Digital and Seagate. M.S. Data Science, Georgetown; B.Eng. Computer Science, NTU Singapore.

PROFESSIONAL EXPERIENCE

PricewaterhouseCoopers - Associate, Data Scientist (Risk Services) — Singapore | Jul 2022 - Jul 2024

- Built a fraud detection system screening **every grant application,** pairing automated document extraction with anomaly models to **cut manual review time by 50%** and surface irregularities that rules-based screening missed.
- Designed and shipped an NLP regulatory analytics pipeline for the **Singapore Exchange (SGX)**, processing filings and disclosures for **every SGX-listed company** into structured compliance signals and replacing line-by-line analyst review.
- Authored technology risk frameworks for AI, cloud, data, and robotics systems, translating model failure modes into controls that legal, audit, and engineering stakeholders could act on jointly.

Envision Digital International - Artificial Intelligence Research Intern — Singapore | May 2021 - Jan 2022

- Developed LSTM, CNN, and TadGAN models for predictive maintenance across a fleet of solar assets, detecting IV-curve faults, PV shading, and irradiance anomalies.
- Deployed the diagnostic models into production inference pipelines used for continuous asset monitoring, and co-authored the resulting paper at the **48th IEEE Photovoltaic Specialists Conference**.

Seagate Technology - Applied AI Engineering Intern — Singapore | Aug 2020- Dec 2020

- Architected a distributed edge-computing framework (Docker, Ray, MXNet) for multi-stream video inference on computational storage.
- Engineered a parallel processing pipeline achieving a **262% throughput improvement** over baseline, and benchmarked AI workload scaling to inform product direction.

SELECTED PROJECTS

HERALD - Claim Verification for AI-Generated Policy Memos · *GPT-4o, Claude, DeBERTa NLI, MCP* — GitHub ↗

- Built a four-tier claim-verification pipeline that routed each claim by a six-type taxonomy through a local DeBERTa NLI model, an LLM-as-judge, a three-persona multi-agent debate, and human review, exiting at the first confident verdict.
- **The pipeline achieved 96% accuracy** on a held-out benchmark, matching a single-call Claude baseline at **23% of the cost.** Diagnosed a prompt-framing error driving spurious escalation and **cut cost a further 86%** with no accuracy loss.
- Shipped as a Next.js and FastAPI app with MCP tool servers over arXiv, World Bank, FRED, GovInfo, and Semantic Scholar, instrumented with Braintrust and OpenTelemetry.

Metro Station Placement via Reinforcement Learning · *PPO, DQN, Stable-Baselines3, Gymnasium* — GitHub ↗

- Built a custom RL environment simulating sequential station placement on the WMATA network, integrating GTFS, LODES, and Census data with Haversine coverage geometry and hourly demand profiles.
- Trained PPO and DQN agents against greedy and k-means baselines to maximize population coverage and minimize passenger wait time.

VibeCheck: Multimodal Restaurant Discovery · *CLIP, Sentence-BERT, FAISS, MLflow, DVC* — GitHub ↗

- Built a cross-modal retrieval system that lets users search restaurants by *ambience* via natural-language description or reference photo, using CLIP and Sentence-BERT embeddings in a shared semantic space.
- Ran dense-vector search with FAISS and clustered venues by aesthetic with UMAP and HDBSCAN; instrumented the lifecycle with MLflow tracking, DVC versioning, and Evidently drift monitoring.

EDUCATION

Georgetown University - M.S. Data Science and Analytics — Washington, D.C. | Aug 2024 – May 2026

- **GPA 4.0 / 4.0.** Graduate Teaching Assistant for six courses, including Machine Learning Foundations, Predictive Analytics, Network Analytics, and Computer Vision & Generative Image Modelling. Data Science & Analytics Returning Student Scholarship.

Nanyang Technological University - B.Eng. Computer Science, Honors (Distinction) — Singapore | Aug 2018 – May 2022

- **GPA 4.45 / 5.0.** Specializations in Data Science and Artificial Intelligence. URECA Research Award. Vice President, NTU Debating Society.

TECHNICAL SKILLS & PUBLICATIONS

Languages & ML: Python, SQL, R, Java, C++, TypeScript, PyTorch, TensorFlow, scikit-learn, Hugging Face, CLIP / Sentence-BERT, FAISS, Stable-Baselines3, OpenCV, Ray / RLlib

Data & Infrastructure: Apache Spark (EC2 clusters), Amazon Web Services (S3, Lambda, Glue, Athena), Microsoft Azure, Google BigQuery, Amazon Redshift, Docker, Kubernetes, FastAPI, MLflow, DVC, Polars, DuckDB, Git

MMethods: Deep learning (convolutional neural networks, recurrent neural networks, transformers), Large Language Model evaluation, Relation-Aware Graph, Natural Language Processing, computer vision, reinforcement learning, time-series forecasting, A/B testing, Bayesian inference

Publication: *Automatic IV Curve Diagnosis with Deep Learning* - 48th IEEE PVSC, 2021. — Paper ↗